

# Hemanth Bhimavarapu AI Analyst

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## SUMMARY

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Accomplished AI Analyst with expertise in Generative AI, Large Language Models (LLMs), and cloud-based AI solutions. Experienced in designing, developing, and deploying AI-driven applications using Python, TypeScript, and Java. I am skilled in fine-tuning and evaluating LLMs (GPT, Gemini) and optimizing ML models for real-world logistics and operational AI systems. Proficient in Google Cloud Platform (GCP), AWS, and MLOps for scalable AI deployment. Adept at data engineering, cloud automation, and CI/CD pipelines to streamline AI workflows. Strong problem-solving skills with a focus on AI research, machine learning, and cloud-native AI architectures.

## TECHNICAL SKILLS

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**Languages & Framework:** Java, JavaScript, TypeScript, Python, React.js, Redux, Html5, Css3, Bootstrap, Spring Boot, Node.js, RESTful APIs, Hibernate, MVC

**Database:** MySQL, PostgreSQL, Microsoft SQL Server, Oracle, MongoDB, Cassandra, Redis

**Tools & Technology:** PromptFoo, AWS (EC2, S3, Lambda, RDS), Azure (App Services, Functions, SQL Database), Jenkins, GitHub Actions, Azure DevOps, JUnit, Mockito, Selenium, Jest, Enzyme, OAuth2, JWT, Secure Coding Practices, Security Audits, Docker, Kubernetes, Maven

**Generative AI & Machine Learning:** LLM evaluation, PromptFoo expertise and test suite creation, Prompt optimization, Automated testing, YAML configuration, Model-graded evaluation, Performance metrics, Data-driven analysis and deploying LLMs (GPT, BERT, T5, Gemini), Model evaluation techniques using Promptfoo, Retrieval-Augmented Generation (RAG).

## EXPERIENCE

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**AI Analyst, Google | GlobalLogic**

**Jan 2025 – present | Remote, USA**

- Analyzed code responses generated by various Large Language Models (LLMs) like GPT, GEMINI, BERT, and others.
- Evaluated the effectiveness, accuracy, and quality of code responses to determine which LLM model performs better under specific conditions.
- Developed and refined prompts using promptfoo to optimize large language model (LLM) outputs, significantly improving unit test pass rates and overall model performance.
- Designed and implemented YAML configuration files for automated LLM evaluation, defining test cases, prompts, and assertion criteria for rigorous model assessment.
- Conducted systematic prompt variations and analysis to identify optimal phrasing and instructions, resulting in enhanced accuracy and adherence to specific output constraints (e.g., word count).
- Created automated testing frameworks utilizing promptfoo to evaluate LLM responses across diverse topics and ensure compliance with defined performance metrics.
- Implemented model-graded evaluations using LLM rubrics within promptfoo to assess output quality and adherence to specific requirements, ensuring high-quality model responses.
- Performed detailed analysis of LLM outputs to identify patterns and areas for improvement, contributing to the development of more robust and reliable AI systems.
- Demonstrated expertise in writing and managing YAML configuration files for complex testing and evaluation environments. Hands on experience with promptfoo, to evaluate LLMs.
- Utilized Cloud SQL and BigQuery for efficient data storage, retrieval, and analytics in high-traffic applications.

- Demonstrated expertise in utilizing PromptFoo for comprehensive LLM evaluation, resulting in significant improvements to unit test pass rates through strategic prompt refinement.
- Wrote clean, maintainable code in Python and TypeScript, ensuring efficient AI system integration.
- Wrote and analyzed code in TypeScript, identifying and fixing bugs and other issues in the software through systematic debugging and troubleshooting techniques.
- Directly increased the pass rate of LLM unit tests by implementing targeted prompt modifications and detailed evaluation methodologies within PromptFoo
- Designed and maintained relational/non-relational databases using MongoDB and PostgreSQL, ensuring data integrity and optimal performance with 99.98% availability.
- Designed and executed automated testing frameworks within PromptFoo, leveraging YAML configurations to define comprehensive evaluation criteria and ensure consistent model performance.
- Successfully implemented LLM rubrics within PromptFoo to perform model graded evaluations, directly contributing to improvements in pass rates by ensuring output adherence to set standards.
- Collaborated with UI/UX designers to translate design mockups into high-quality code, ensuring a consistent look and feel across the application.

### **Software Engineer, United Parcel Services**

**Sep 2023 – Feb 2024 | New Jersey, USA**

- Configured Jenkins pipelines to automate build, test, and deployment processes, reducing manual effort and deployment time.
- Implemented Terraform scripts to provision and manage cloud infrastructure resources efficiently.
- Ensured code quality and security compliance by integrating SonarQube into the CI/CD pipeline for static code analysis.
- Developed Jenkins pipeline scripts to streamline CI/CD workflows, reducing deployment time and increasing reliability.
- Conducted static code analysis with SonarQube, ensuring adherence to best practices, security standards, and code quality.
- Optimized pipeline performance by implementing parallel execution strategies, reducing build times by 30%.
- Participated in various stages of the Software Development Life Cycle (SDLC), encompassing requirements gathering, modeling, analysis, design, development, and testing.
- Developed application components in an agile environment, employing Test-Driven Development (TDD) to ensure efficient and reliable quality of code.
- Designed and implemented RESTful APIs using Spring Boot for transaction processing, user management, and product services, including authentication and authorization features using Spring Security (JWT and OAuth).
- Integrated APIs into the front-end via Axios and managed API testing and documentation using Postman, ensuring robust endpoint functionality.
- Implemented advanced React features such as Hooks and Context API to manage global state without external libraries, reducing bundle size and improving load times.
- Utilized JIRA for sprint tracking and issue resolution, maintaining a clear record of development progress and backlogs.
- Deployed applications using Docker containers, managed through Kubernetes clusters on AWS EC2 instances, automating deployment processes and scaling operations.
- Leveraged Google Cloud SDKs, Pub/Sub, Cloud Storage, and BigQuery to streamline data processing and analytics workflows.
- Integrated Google Cloud Logging and Monitoring (Stackdriver) to provide real-time system health insights and proactive alerting.

- Created and maintained RESTful APIs to support various client applications, ensuring high reliability and ease of integration using Java and react JS.
- Managed deployment and scaling of applications using Azure App Services and Azure Kubernetes Service (AKS), enhancing scalability and performance.
- Developed comprehensive unit and integration tests with JUnit and Mockito, ensuring 95% code coverage and minimizing bugs in production.
- Designed and implemented event-driven architectures using Google Cloud Pub/Sub and Cloud Functions, improving real-time data processing.
- Developed and maintained RESTful APIs in Java to interact with Google Cloud services, ensuring seamless data flow across applications.
- Analyzed existing Python services and rewrote them in Java, ensuring functional parity while improving performance and maintainability.
- Conducted code refactoring and optimization while converting Python services into Java, reducing execution time and memory usage
- Optimized data pipelines by refactoring Python-based ETL processes into Java-based solutions on Google Cloud.
- Collaborated with cross-functional teams to validate migration efforts and optimize service performance.
- Set up monitoring and logging with Azure Monitor and Application Insights, achieving 99.9% application uptime and quick issue resolution.
- Conducted performance tuning to enhance application response times, resulting in a 25% improvement in user experience.
- Managed version control with Git and performed continuous integration and deployment (CI/CD) using Jenkins and Docker.
- Set up and monitored application performance using Grafana and Prometheus, allowing real-time monitoring of system health and proactive troubleshooting of issues.
- Implemented security measures in application development such as HTTPS for secure data transmission, and bcrypt for hashing and salting passwords stored in MongoDB.
- Designed and executed comprehensive API tests using Postman to validate RESTful services, ensure endpoint security, and significantly reducing bugs and improving system reliability before deployment.
- Designed and optimized relational database schemas in MySQL, ensuring data consistency and high performance.

## **EDUCATION**

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**Texas A&M University**  
Master's in Computer Science

**August 2022 – May 2024**  
Kingsville, USA